Comprehensive Research Overview: IIT Bombay Aerospace Engineering Faculty – Current and Recent Research Activities

This exhaustive list collates the most **specific, current** research thrusts of all Aerospace Engineering faculty at IIT Bombay (2021–2025), detailing fine-grained topics suitable for student collaboration.

Prof. Shashi Ranjan Kumar: nonlinear dynamics and control of aerospace vehicles, development of hardware-in-loop inertial navigation systems, design and fabrication of solar-powered miniature aircraft, dynamics and stability analysis of fixed-wing micro air vehicles, precision MEMS sensor development for trajectory estimation;
Prof. Abhijit Gogulapati: computational aeroelasticity and aerothermoelasticity of composite wings, reduced-order modeling of fluid–structure interaction, design optimization of flutter-suppression control laws, morphing wing structures via active materials;
Prof. Aniruddha Sinha: parabolized stability analysis and wavepacket modeling for high-speed jet noise mitigation, reduced-order models of supersonic jet mixing controlled by plasma actuators, feedback control strategies for airfoil separation delay, aeroacoustic source identification via modal decomposition of large-scale jet datasets;
Prof. Arnab Maity: adaptive feedback control and robust stability for multivariable aerospace systems, model predictive control for small-satellite attitude dynamics, data-driven system identification of launch-vehicle structural modes;
Prof. Avijit Chatterjee: high-fidelity CFD for radar cross-section prediction of stealth intakes, time-domain finite-element electromagnetics for aircraft signature reduction, integrated aerodynamic–EM optimization of flying-wing configurations;
Prof. Bhupendra Khandelwal: fatigue life prediction of composite fan blades under combined loading, high-strain-rate constitutive modeling of polymer-matrix composites, environmental degradation modeling of wind-turbine blade materials;
Prof. Chandra Sekhar Yerramalli: multiscale fatigue and damage modeling in fiber-reinforced composites, modeling of combined loading fatigue failure in aero-structural components, wind-turbine blade aeroelastic performance under dynamic gust loads;
Prof. Dhwanil Shukla: low-Reynolds-number aerodynamics of unmanned rotorcraft, flow-diagnostic PIV/LDV techniques for helicopter wake interactions, drone airframe design and rotor-craft-scale aerodynamic optimization;
Prof. Ganapathi Bhat: object-oriented thermal systems modeling for avionics cooling, conjugate heat-transfer analysis of aircraft thermal management systems, transient thermal response of electronic assemblies in aerospace environments;
Prof. Hemendra Arya: higher-order zigzag and layerwise models for vibration analysis of laminated composite beams and plates, experimental impact behavior of fabric composites, development of MEMS-based sensors for structural health monitoring of aerospace structures;

Prof. Hrishikesh Gadgil: atomization and breakup modeling of aeroengine fuel sprays, effervescent and pressure-swirl atomizer design for alternative fuels, acoustic forcing effects on liquid fuel atomization in combustor injectors;
Prof. J. C. Mandal: all-Mach CFD formulations for incompressible, compressible, hypersonic and multiphase flows, high-resolution unstructured-grid solvers for space-vehicle aerodynamics, semi-analytic methods for fluid–structure interaction in launch-vehicle components;
Prof. Kowsik Bodi: magnetohydrodynamic flow simulations for electric propulsion and liquid-metal divertors, arc-plasma modeling in wind-tunnel facilities, reacting flow simulation for thruster and plasma‑wind-tunnel coupling;
Prof. Krishnendu Haldar: constitutive modeling and finite-element simulation of magnetoactive polymers and magnetic shape-memory alloys, bifurcation and stability analysis of field-induced phase transformations, rate-dependent anisotropic viscoelastic modeling for soft biological tissues;
Prof. Krishnendu Sinha: shock–turbulence and shock–boundary-layer interaction modeling in hypersonic flows, Reynolds-stress turbulence modeling for high-enthalpy air chemistry, scramjet intake and re-entry flow simulations with chemical nonequilibrium;
Prof. Niranjan Krishna Naik: thermomechanical impact modeling of woven and braided composites under low-velocity and ballistic loads, strain-rate dependent behavior of polymer–matrix composites, permeability and resin-transfer molding characterization for advanced composite sandwich structures;
Prof. P. J. Guruprasad: discrete dislocation dynamics modeling of creep and plasticity in nickel-based superalloys, development of shape-memory polymer composites for morphing aircraft skins, variational asymptotic homogenization of auxetic lattice metamaterials;
Prof. Prabhu Ramachandran: meshless SPH methods for compressible and incompressible Euler flows, entropically damped artificial compressibility schemes, object-oriented DSMC codes for rarefied flow around complex aerospace geometries;
Prof. Pradeep A. M.: transonic flow feature identification in axial compressor rotors, unsteady secondary flow characterization near stall inception, inter-spool duct aerodynamics under compressor-stage distortion, three-dimensional blade-tandem design optimization using phase-locked ensemble averaging;
Prof. R. K. Pant: design, fabrication and testing of lighter-than-air mooring masts and indoor/outdoor airships, branch-and-bound ground-movement routing algorithms for airport optimization, transient heat-transfer modeling in stratospheric airships;
Prof. Rohit Gupta: geometric mechanics and control theory for aerospace vehicles, optimal control frameworks for orbit transfer under perturbations, symmetry-based reduction techniques for spacecraft attitude dynamics;
Prof. Sudarshan Kumar: microcombustion and flameless combustion in micro-reactor channels, measurement of laminar burning velocities at elevated temperatures, modeling of supersonic and MILD combustion regimes in rocket and gas-turbine engines, active instability control via pulsed detonation actuation;
Prof. T. Chandra Sekar: experimental aero-servo-elastic and aerothermoelastic testing, fluidic thrust-vectoring nozzle performance mapping, active flutter suppression in gas-turbine blade cascades, combustor–nozzle integration for performance and emission optimization;
Prof. V. Menezes: miniature shock-driven microjet injectors for needle-free drug delivery,

scramjet engine flow-field thermal and inertial parameter characterization, high-speed free-surface shock–water interaction studies;

Prof. V. Venkateswara Rao: advanced experimental methods for compressible combustion diagnostics, flame-stabilization studies in combustion chambers, mixing enhancement techniques in supersonic combustors;

Prof. Vineeth Nair: Lagrangian coherent-structure analysis of combustion noise sources, multifractal precursors to aeroelastic flutter, data-driven reduced-order models for thermoacoustic instability prediction;

Prof. Nagendra Kumar: combustion chemistry and deflagration limit studies of ammonium perchlorate monopropellants, binder influence and rapid depressurization effects on composite solid-propellant combustion.